Panel Augmented Macroeconomic Data and Recurrent Neural Networks Forecasting

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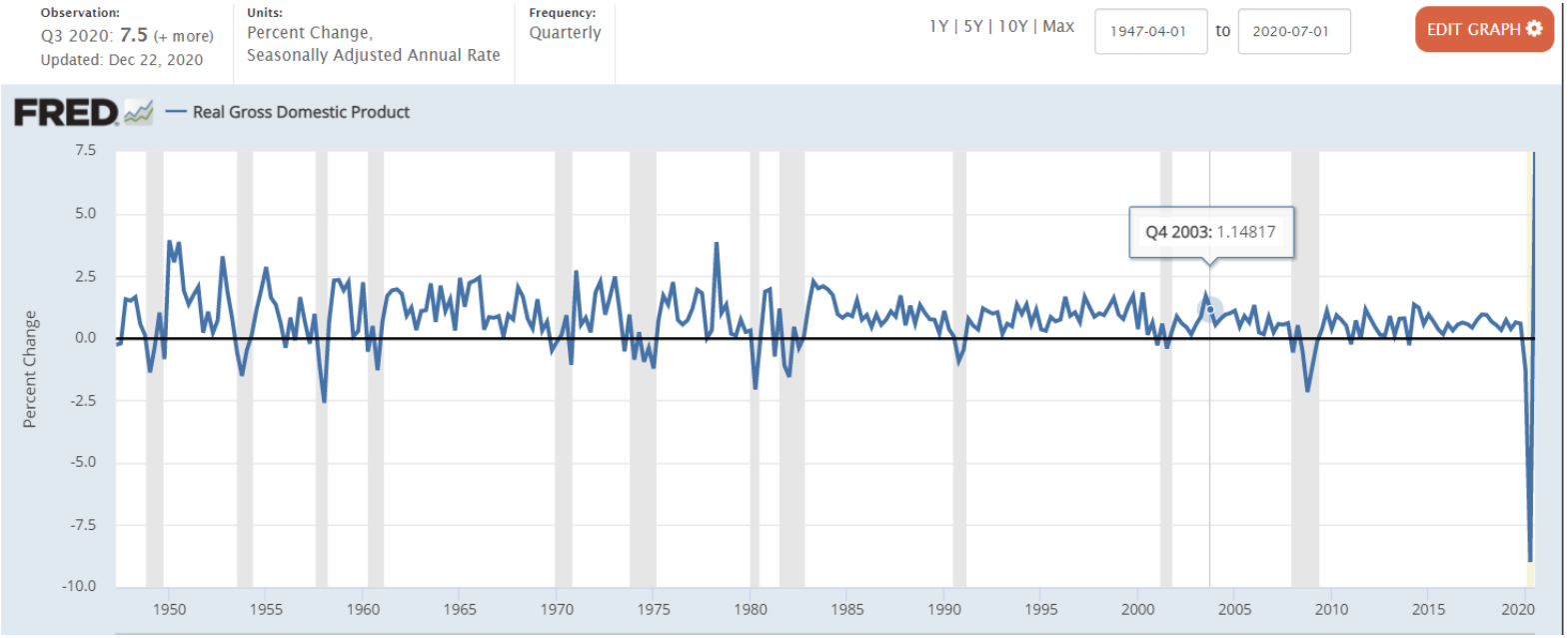
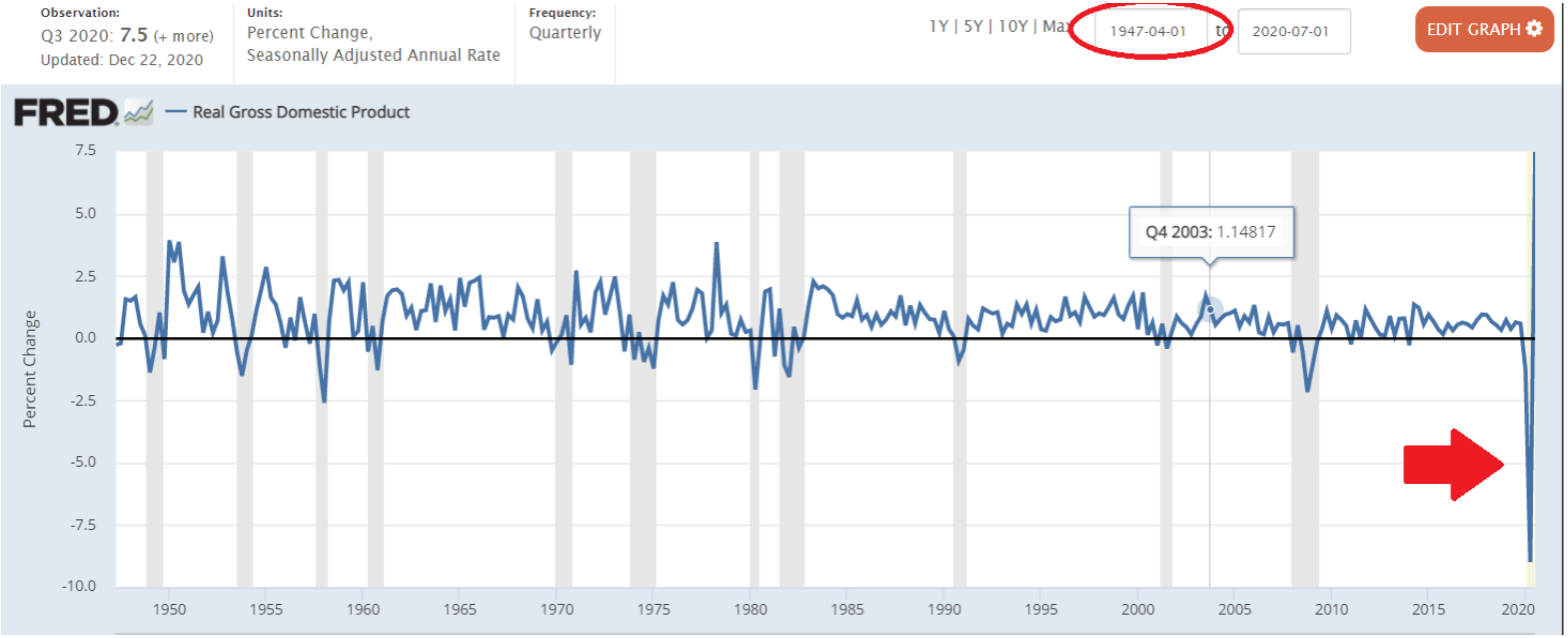
January 22nd, 2020

Motivation: GDP Data



Motivation: GDP Data



Central Question: How can we augment macroeconomic data to improve the external validity of our models?

- **Augment the size of our data**: We advocate fixing this dearth of data problem by using a panel of 49 other countries + US data
- **External Validity**: We offer evidence that using a panel dimension when estimating models improves forecasting performance in a wide variety of circumstances
- **More fundamental conclusions**: We demonstrate that this augmented data makes reduced form (and structural) models more policy invariant
- **Flexible models**: We show that this procedure opens up the use of more flexible/non-parametric models like our 18,000 parameter recurrent neural network that outperforms all baselines in forecasting

The Data and Baseline Models

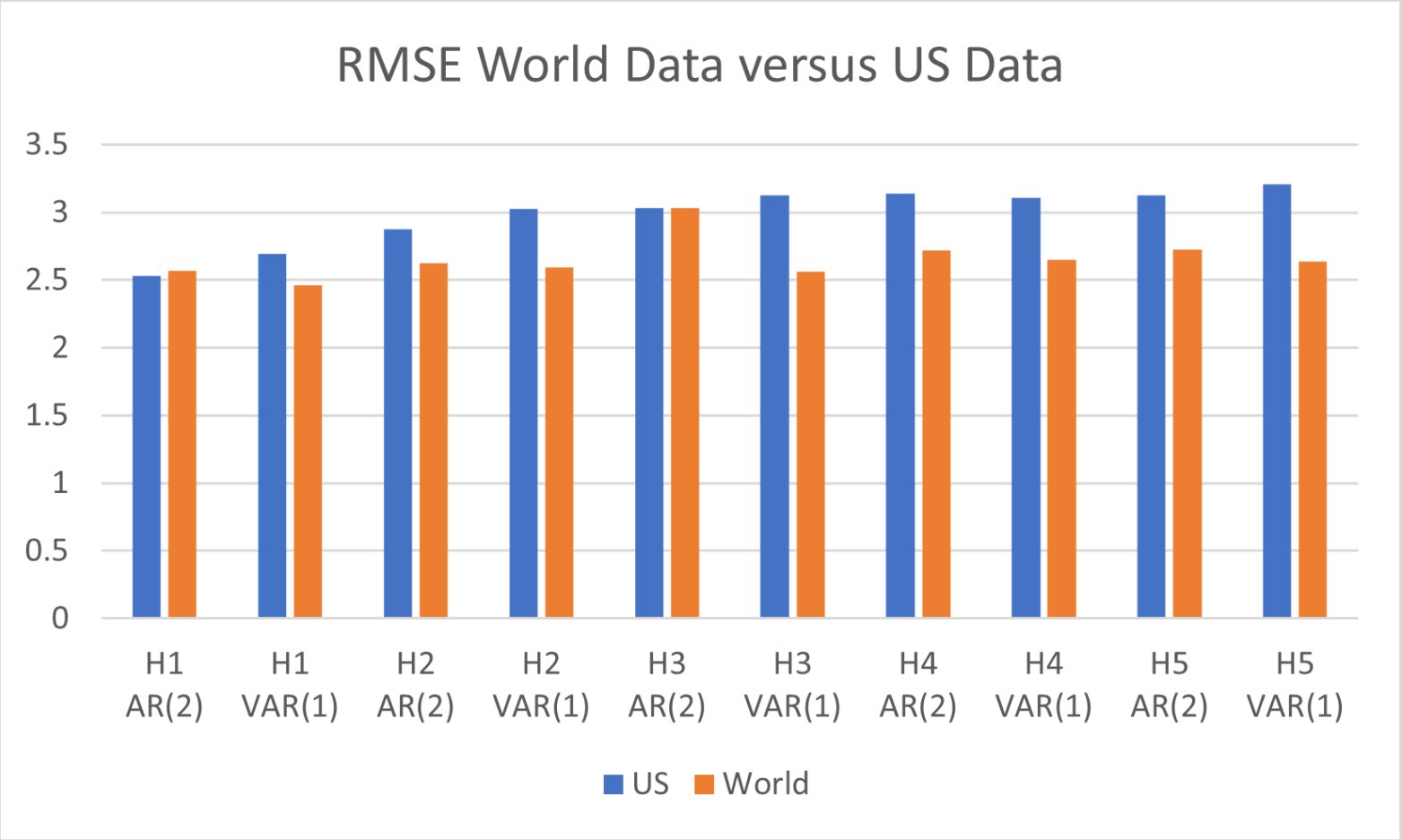
- We evaluate the performance of our models via GDP forecasting
- We use growth rate data from 50 different developed countries (GDP, consumption, unemployment) to forecast GDP growth
- Data Sources: World Bank and Trading Economics via Quandl
- Models:
  1. AR(2)
  2. VAR(1)
  3. Smets Wouters 2007 DSGE
  4. Factor Model*
  5. Recurrent Neural Network

The Baseline Models

- Baseline Models
  - AR(2) model: forecasts GDP with two lags in a linear manner
  - VAR(1) and VAR(4) models: forecasts GDP using one lag from our three independent variables
  - DSGE model: A structural model that uses economic theory to forecast
  - Factor model: Uses large cross section of data (248 variables), condenses the information down into lower dimensional factors via PCA, and uses a linear model to regress GDP on the factors and a GDP lag

Appendix: Model Description

Forecasting Improvement



Back to Model Reliability

Improvement Not Obvious A Priori

- More data does not always lead to better forecasting performance
  - Literature on negative transfer, see (Wang 2019) among many
- The new data is not a representative of US GDP performance as it is a different country–biased
- However the increase in data will also reduce variance of our models

Forecasting Performance Not Constrained to US

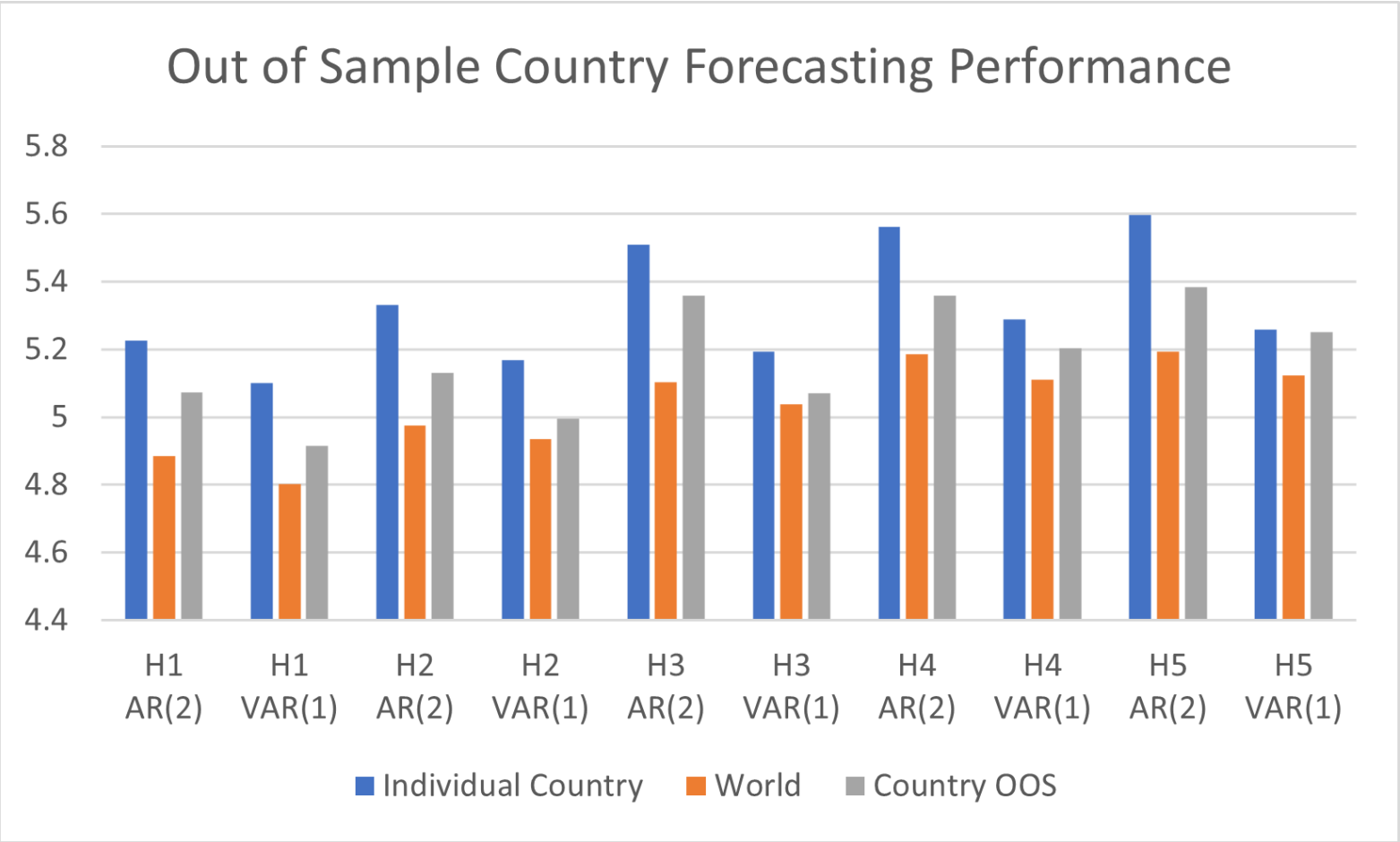
Table 1: Average Forecasting Performance

Time (Q's Ahead)	1Q	2Q	3Q	4Q	5Q
AR(2)					
Local Data	5.1	5.1	5.3	5.4	5.4
World Data	4.8	4.8	5.0	5.1	5.1
VAR(1)					
Local Data	5.0	5.0	5.4	5.4	5.5
World Data	4.7	4.8	4.9	5.0	5.0
VAR(4)					
Local Data	8.4	7.2	8.0	7.8	8.8
World Data	4.7	4.8	4.9	5.0	5.0

Cross Country Interpretation

- The table shows the use of cross country data improves forecasting performance on average across all 50 countries
- Using the cross country data allows the model to learn more fundamental parameter values that generalize across many countries and policy regimes
  - The panel approach is a single model trained on world panel data that makes forecasts for 50 different countries

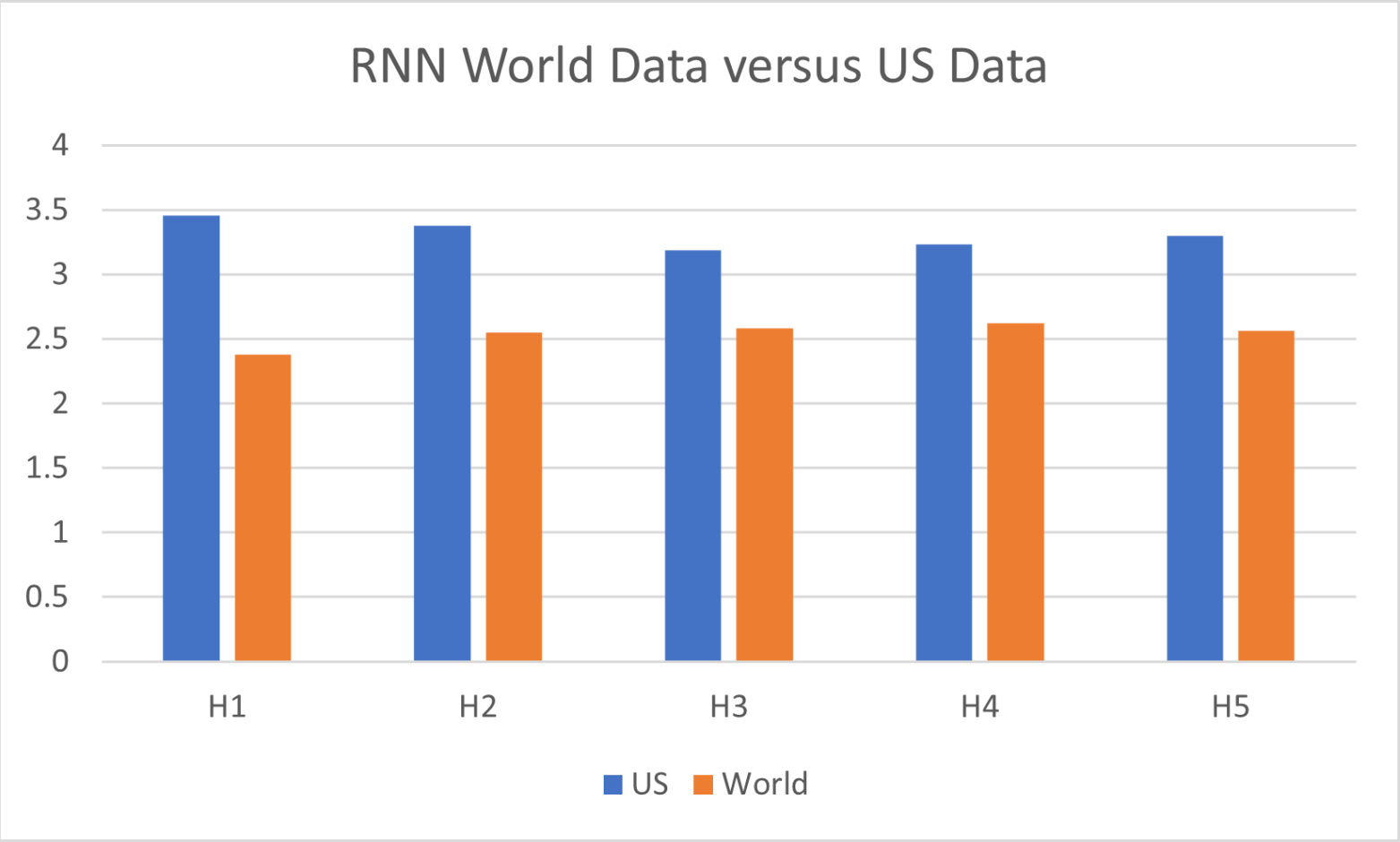
Policy/Country Invariance using Fully Out-of-Sample Predictions



Invariance Across Countries

- This table show forecasting performance only using local data (first bar in groups), using world data (second), and using all country data except the in-sample country being forecasted (third)
- Performance is aggregated across all 50 countries
- In all situations, pooled except the in-sample country outperforms using only in-sample country to forecast
- On average having pooled training data except for the in-sample country is resposible for more than 50 percent of the reduction in RMSE gained from moving from only in-sample data to the full panel
- Provides strong evidence that using panel augmentation improves the policy invariance and generalization to unseen countries for reduced-form models
- Next I will discuss how this panel augmentation scheme can improve the quality of model that can be estimated by training a 18000 parameter recurrent neural network

RNN Forecasting Improvement



Our Model

- We use a Recurrent Neural Network (RNN) as our forecast model
- An RNN is a state space model
- Like a linear state space model, the RNN will have latent states similar to the below representation of the linear state transition equation:

$$h_t = Ah_{t-1} + B + \epsilon \tag{1}$$

- ...and a measurement equation

$$y_t = Ch_t + Dx_t + E + \epsilon \tag{2}$$

- However, the main innovation of an RNN is to use gates, a logistic regression that is element-wise multiplied to the states (next slide)
  - Conditioned on previous information, the model can control the states allowing the eigenvalues of A to be arbitrary

A GRU Cell

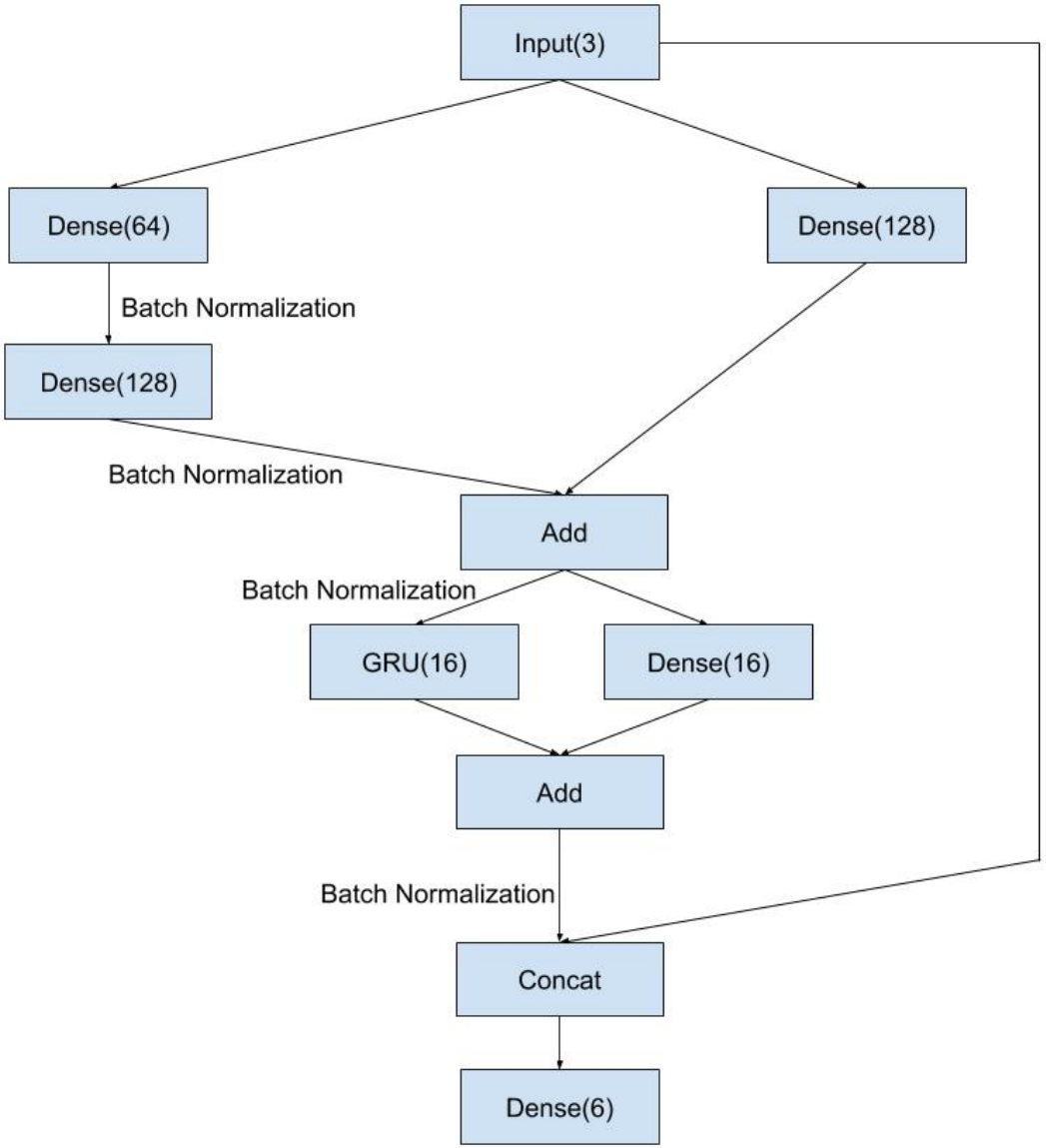
State Transition Equations:

$$
\begin{aligned}
z_t &= \sigma_g(W_z x_t + U_z h_{t-1} + b_z) \\
r_t &= \sigma_g(W_r x_t + U_r h_{t-1} + b_r) \\
\hat{h}_t &= \phi_h(W_h x_t + U_h(r_t \odot h_{t-1}) + b_h) \\
h_t &= (1 - z_t) \odot h_{t-1} + z_t \odot \hat{h}_t
\end{aligned}
$$

Measurement Equation:

$$
y_t = W_y x_t + U_y h_t + b_y
$$

Architecture Structure

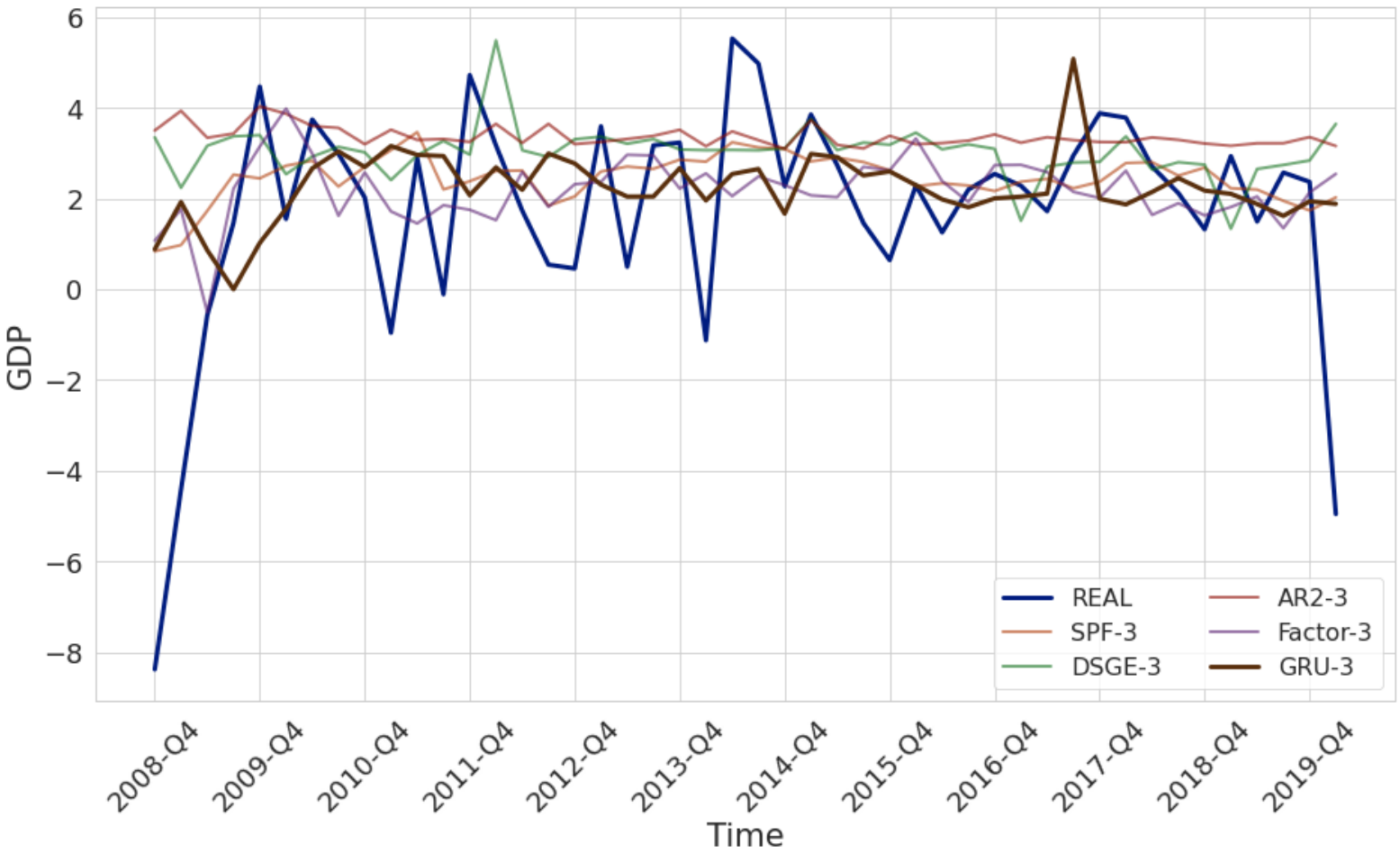


GRU Root Mean Squared Error

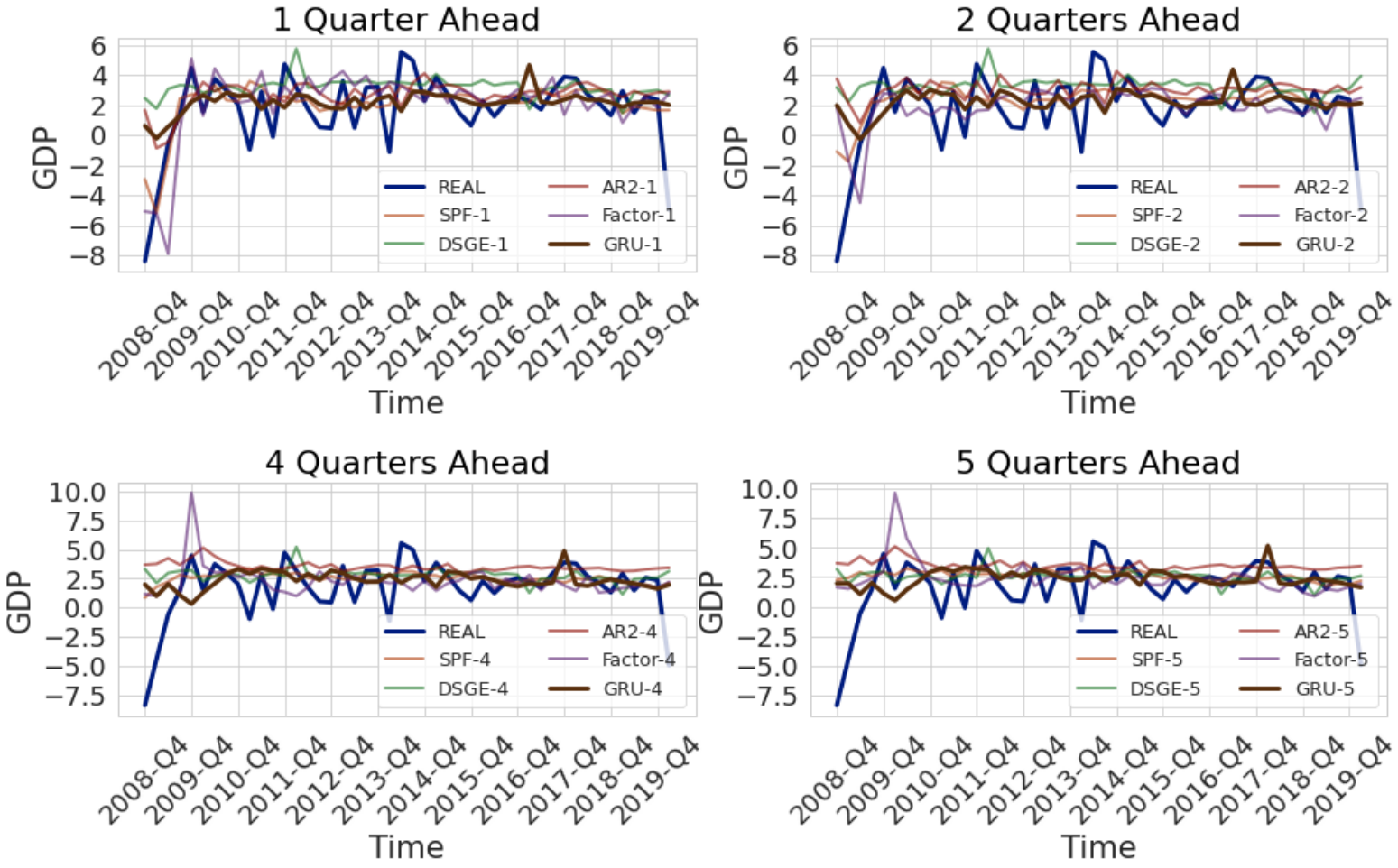
Time (Q's Ahead)	1Q	2Q	3Q	4Q	5Q
VAR(1)					
US Data	2.9	3.8	4.4	4.5	4.7
World Data	2.5	2.9	3.0	3.1	3.1
AR(2)					
US Data	2.5	2.9	3.0	3.1	3.1
World Data	2.6	2.6	2.7	2.7	2.7
Smets Wouters DSGE					
US Data	2.8	3.0	2.9	2.8	2.7
Factor					
US Data	**2.2**	**2.5**	2.5	2.7	2.9
GRU					
Best	2.4	2.5	2.6	**2.6**	**2.6**
Mean Forecast	2.3	2.5	**2.5**	**2.6**	**2.6**
Median Forecast	2.3	2.5	**2.5**	**2.6**	**2.6**
SPF Median	1.9	2.1	2.4	2.5	2.7

*All GRU models use entire world panel

Graph of Forecast Performance: 3 Quarters Ahead



Graph of Forecast Performance: Tile



Out of Sample Forecast Comparison: Expansions

Time (Q's Ahead)	1Q	2Q	3Q	4Q	5Q
VAR(1)					
US data	2.3	2.6	2.9	3.0	3.0
World data	2.1	2.2	2.2	2.2	2.2
AR(2)					
US data	1.7	1.7	1.8	1.9	1.9
World data	**1.6**	1.6	1.6	**1.5**	**1.5***
Smets Wouters DSGE					
US data	1.8	1.8	1.7	1.6	1.5*
Factor					
US data	1.6	**1.6**	**1.6**	1.9	2.1
GRU*					
Best	1.8	2.3	2.0	2.0	1.9
Mean Forecast	1.7	1.7	1.7	1.7	1.7
Median Forecast	1.7	1.7	1.7	1.7	1.7
SPF Median	1.4	1.5	1.5	1.5	1.5

*All GRU models use entire world data panel

Out of Sample Forecast Comparison: LSTM Model

Table 2: LSTM Forecast Performance

Time (Q's Ahead)	1Q	2Q	3Q	4Q	5Q
Best RMSE	2.4	2.6	2.6	**2.6**	**2.6***
RMSE of Mean	2.4	2.6	2.5	**2.6**	**2.6***
RMSE of Median	2.4	2.5	2.5	**2.6**	**2.6***
Mean RMSE	2.4	2.6	2.5	**2.6**	**2.6***
Std Dev RMSE	0.05	0.07	0.05	0.06	0.06

Appendix: Additional Results

Does our RNN learn economically Relevant Information?

- In the process of testing whether the states in our RNN are correlated with a panel of economic variables
- Some circumstantial evidence the model is learning something correlated to things like total wealth
- Hopefully in a month we'll be able to show you a correlation table highlighting economic variables most correlated with our states

Conclusion

- We showed how one can use data augmentation to improve the policy invariance of these models
- Our neural network forecasting model was competitive with the best economic forecasting models
- With the suggestive evidence of policy invariance, in future work, we plan use these models for counterfactual analysis

Thank You

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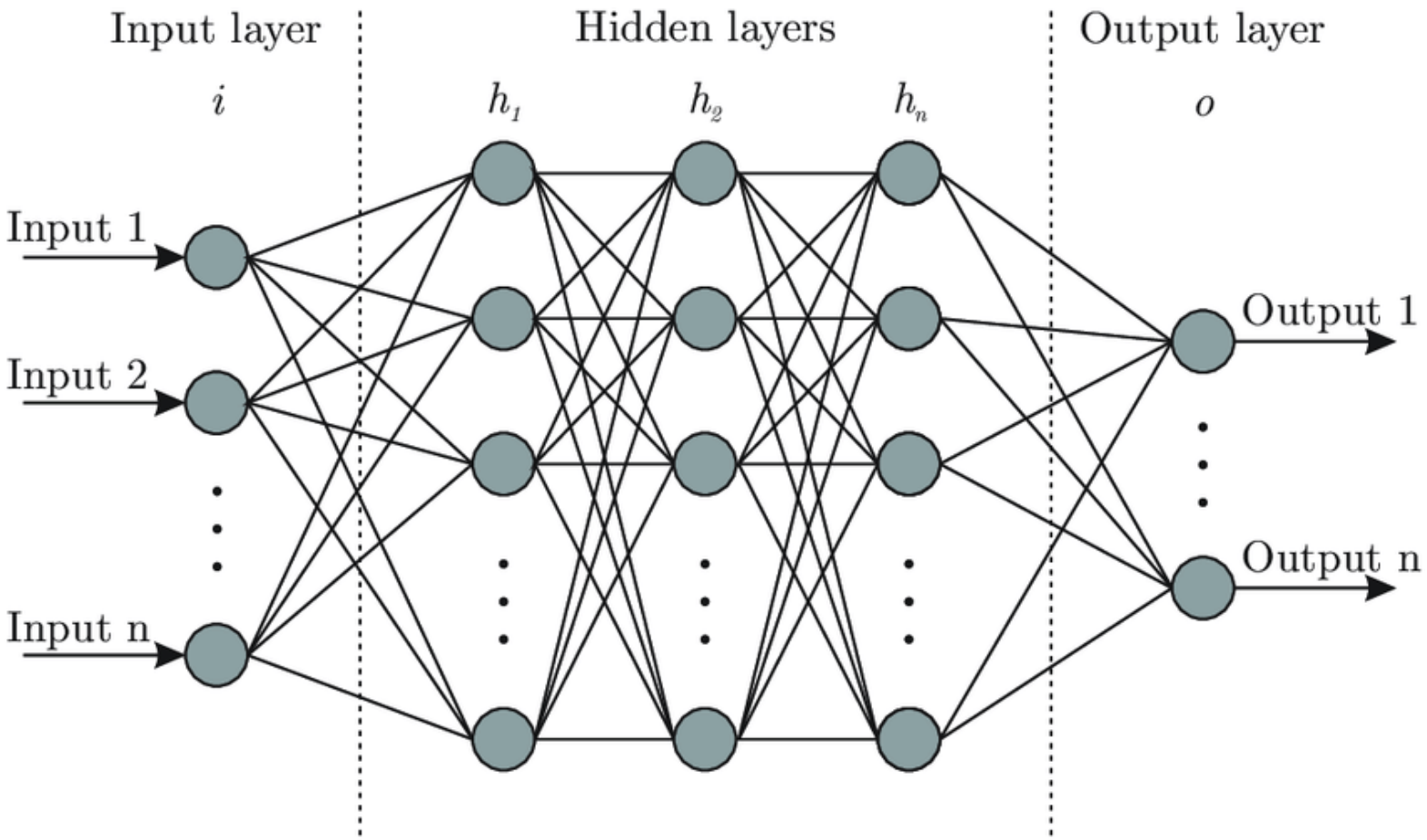
Appendix

Dense Layers: A Picture



Dense Layers

- A dense layer in a neural network is simply a vector regression $y = \sigma(\beta x)$ with a link function (called an activation), σ, which makes the model nonlinear
- y is a vector and so β is a matrix
- y becomes the multivalued input of the next layer, ie

$$y_3 = \sigma(\beta_3(y_2)) = \sigma(\beta_3(\sigma(\beta_2 x))) \tag{3}$$

- The activations σ are essential because a linear combination of linear transformations is still linear so without the activations the model doesn't become more expressive

Dense Layers: An Example

- Our input, x, is a 3 dimensional vector (GDP, consumption, unemployment) with 250 time-steps
- Like in logistic regression, x is input into the first layer: $y_2 = \sigma(\beta_2 x)$
- y_2 is now the input, like x, into the second layer
  - If we want y_2 to be size 128, then by definition β_2 is 128 x 3
- If,

$$y_3 = \sigma(\beta_3 y_2) = \sigma(\beta_3(\sigma(\beta_2 x))) \tag{4}$$

  and y_3 is the 1 dimensional output, then since y_2 is 128 x 250 then β_3 is 1 x 128
- In this case, y_2 is the only hidden layer, but you can imagine having many more hidden layers before producing an output

Back to Model Architecture

Baseline Models: AR(2)

- We use a linear model which has two lags of GDP to forecast ahead
- Despite the simplicity, this is one of the workhorse models among forecasting practitioners (Hamilton 1994)
- We find that the AR(2) outperforms the Smets Wouters DSGE at shorter time intervals (1-2 quarters ahead), but is outperformed by Smets Wouters at longer intervals (4-5 quarters ahead)
- A factor model augmented with one GDP lag seems to dominate the AR(2)

Back to Baselines

Baseline Models: DSGE (Smets and Wouters 2007)

- Smets Wouters is New Keynesian DSGE model that is essentially an extension of the Christiano et. al. 2005 model estimated in a Bayesian framework
- The model is geared towards forecasting rather than macroeconomic analysis
- Despite limited attention paid by practitioners, we think this model deserves more attention, especially at longer time horizons
- The model outperforms AR(2) and factor models at longer horizons, but is unable to detect the great recession at shorter horizons which leads to under-performance

Back to Baselines

Baseline Models: Factor Models

- These models were introduced by Stock and Watson 2002 and Forni et. al. 2000
- These models take a large cross section of data (in our case 248 data series) in order to forecast and PCA regression, in our case, reduce the large cross section to 8 factors
- We extend the factor model highlighted in Fred-QD (McCraken and Ng 2020) by combining factors estimated in a pseudo-out-of-sample manner along with a lag of GDP for forecasting
- This model is the most formidable competitor to the neural network, however, it under-performs over long horizons likely because of variance issues and the limited predictive power of the cross section of variables

Back to Baselines

Model Bias and Variance

Table 3:

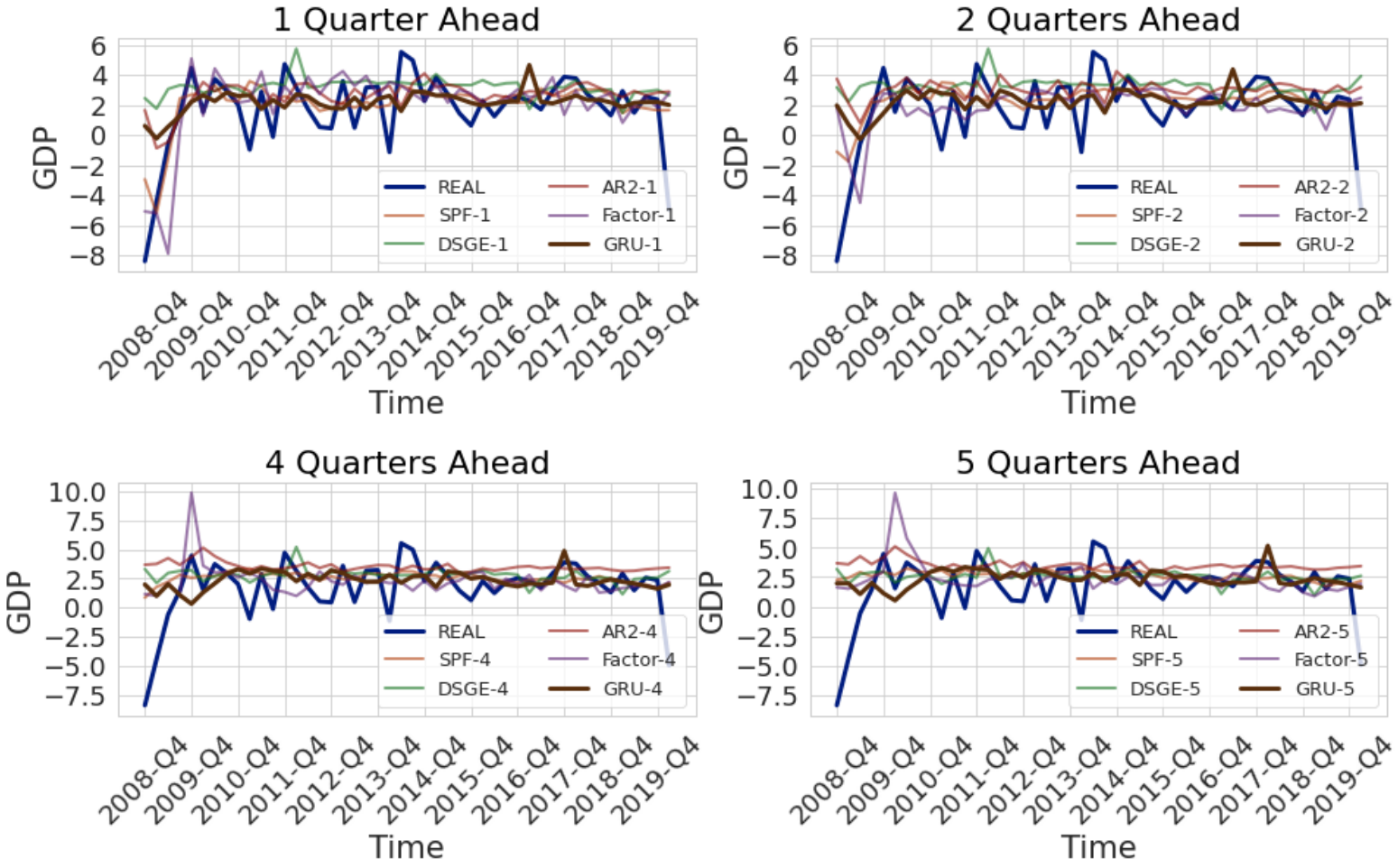
	Forecast Bias				
	(1-Qtr)	(2-Qtrs)	(3-Qtrs)	(4-Qtrs)	(5-Qtrs)
GRU	0.459*	0.480*	0.506*	0.620*	0.644**
	(0.343)	(0.369)	(0.365)	(0.379)	(0.375)
SPF	0.331	0.600**	0.723**	0.804**	0.901***
	(0.274)	(0.302)	(0.335)	(0.347)	(0.372)
DSGE	1.459***	1.513***	1.300***	1.058***	0.827***
	(0.354)	(0.378)	(0.544)	(0.386)	(0.385)
AR2	0.937***	1.317***	1.636***	1.795***	1.780***
	(0.351)	(0.381)	(0.380)	(0.383)	(0.384)
Factor	0.432*	0.163	0.459	0.533*	0.699**
	(0.328)	(0.449)	(0.367)	(0.390)	(0.414)

Notes: ***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Reliability of Data Trained on US data vs World Data

Table 4: GRU Monte Carlo Simulations

Time (Q's Ahead)	1Q	2Q	3Q	4Q	5Q
GRU World Data					
Mean RMSE	2.4	2.6	2.5	**2.6**	**2.6***
Std Dev RMSE	0.06	0.06	0.05	0.06	0.06
GRU US Data					
Mean RMSE	2.6	2.8	2.7	**2.7**	2.8
Std Dev RMSE	0.34	0.34	0.33	0.24	0.44

Back to World vs US Data Forecasting Performance Table Back to Results

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